

Understanding users' news-sharing behaviors: roles of risk perception, believability, fake news awareness and social tie variety

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Abstract

Purpose – This study aims to investigate the impact of news quality on users' risk perceptions toward online news and its subsequent influence on perceived believability and user engagement in sharing news. Additionally, we explore the moderating effects of fake news awareness and social tie variety.

Design/methodology/approach – Drawing upon the social amplification of risk framework, this study investigates the relationship between news quality and users' news-sharing behaviors, along with its underlying mechanism. An online questionnaire involving 399 eligible participants was employed for hypotheses testing, and the structural equation model served as the main analytical method.

Findings – The influence of news quality on users' news-sharing behavior is sequentially mediated by risk perception and perceived believability. Individuals with a heightened awareness of fake news or a diverse social tie are more inclined to perceive greater risks associated with news-sharing behavior and question news authenticity.

Originality/value – This study contributes to the existing literature on users' news-sharing behaviors by examining the influence of risk perception on the relationship between news quality, perceived believability and users' news-sharing behavior. Additionally, it explores the moderating effects of fake news awareness and social tie variety. Our findings offer valuable insights into comprehending user inclinations towards news sharing and mitigating the dissemination of fake news.

Keywords News quality, Risk perception, Fake news awareness, Social tie variety,
Social amplification of risk framework, Sharing behavior

Paper type Research paper

1. Introduction

On the Internet, users are provided with convenient access to online news and seamlessly engage in sharing and discussing it with others. As user engagement increases, there has been an exponential growth in the volume of online information. However, internet regulators lack sufficient resources to monitor the quality of all information. Consequently, online users

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encounter a plethora of low-quality news content, including attention-grabbing headlines and even fabricated videos. The quality of news has emerged as a pivotal determinant influencing users' inclination to share news, particularly in the online environment inundated with fake news (Wang *et al.*, 2022; Wu *et al.*, 2022).

The proliferation of such low-quality online news engenders adverse repercussions for those who disseminate them (Dennis *et al.*, 2023). Numerous cases serve as exemplars of the potential risks associated with low-quality online news, thereby indicating that the presence of low-quality content can act as an indicator for users' potential losses from such specific instances of substandard news. For instance, unhealthy diet or fitness advice is widely circulated on various platforms without scientific validation, promoting extreme weight loss methods or unproven fitness programs. If users believe and adopt these suggestions without caution, they may jeopardize their health inadvertently. In one scenario, an attacker impersonated a bank via email claiming that the user's account had a security issue requiring verification by clicking on a link; unknowingly, users clicked on the link and divulged their bank account details resulting in funds being stolen from their accounts. Additionally, false news stories about new vaccines that can supposedly cure cancer have been disseminated through social media platforms; when users unquestioningly forward such messages without verifying their authenticity first-hand not only misleads the public but also encourages people to disregard formal medical advice thereby posing potential health threats.

Despite the recognition of risks associated with sharing low-quality news, our understanding of the relationship between news quality and users' news-sharing behaviors remains limited in terms of the role played by risk perceptions. Previous studies have examined how news quality directly influences users' news-sharing behavior and explored its mediation through perceived believability (Kumar *et al.*, 2023) and perceived usefulness (Bhagat and Kim, 2023), but they have overlooked the significance of risk perceptions (Lin and Wang, 2020). Disseminating high-quality news not only provides recipients with valuable and timely information but also enhances senders' status, reputation, and image (Zheng *et al.*, 2013). Conversely, when disseminating low-quality news, spreaders may face various risks such as a decline in personal social status (Thompson *et al.*, 2020) or reputational harm (Bhagat and Kim, 2023). This implies that losses incurred by spreaders from sharing low-quality news are not confined to potential losses related to specific pieces of information being spread; instead, they may experience an amplification of risks within the highly interconnected internet.

The significance of risk perception in users' news-sharing behavior can be elucidated by the Social Amplification of Risk Framework (SARF), which is designed to encompass factors relevant to public reactions towards specific risks or risk events (Bearth and Siegrist, 2022). News quality serves as a risk indicator that influences users' perception of risk associated with low-quality news, and subsequent behavioral responses. According to SARF, risk amplification occurs in two stages: during the transmission of risk information and through social responses (Kasperson *et al.*, 1988). In the transmission stage, the highly interconnected internet may amplify the perceived risks as spreaders may worry about reputational losses and social status implications. Subsequently, when individuals perceive heightened levels of risk, their perceived believability towards all online news diminishes (Bearth and Siegrist, 2022), suggesting that risk perception may overshadow perceived believability (Seyd and Bu, 2022) thereby impacting users' news-sharing behavior (Schneider *et al.*, 2021). Therefore, drawing upon the SARF, this study incorporates risk perception to analyze the underlying mechanism and poses the following research question: *What role does risk perception play in shaping the relationship among news quality, perceived believability, and users' news-sharing behavior?*

The SARF focuses on how low-quality characteristics contribute to users' risk perception through risk amplification stations (Wirz *et al.*, 2018). The efficacy of risk amplifications is significantly affected by information dissemination channels and user characteristics. The significant consequence of low-quality news is attributed to the highly interconnected social network on the Internet, which varies among users with different levels of engagement in social

media, such as their frequency of engagement in online discussions (Zhang and Cozma, 2022) and exposure to fact-check information (Ho *et al.*, 2022). Given our focus on news quality amidst the widespread dissemination of fake news on the Internet, users' awareness about the presence of low-quality or fake news is directly relevant to their risk perceptions. Recognizing the existence of fake news enhances users' skepticism towards encountered information while encouraging active participation in verification endeavors (Torres *et al.*, 2018). When users adopt a more skeptical mindset towards news, they recognize higher levels of risk and uncertainty associated with sharing such content, leading them to refrain from doing so (Omar *et al.*, 2024). While acknowledging the moderating role of fake news awareness in the relationship between news quality, users' evaluations of perceived believability, and their sharing behavior, our understanding regarding its influence on the association between news quality and risk perception remains limited. Therefore, we propose our second research question: *How does fake news awareness moderate the effect of news quality on users' news-sharing behavior, mediated by risk perception and perceived believability?*

As depicted in the SARF, the information dissemination channel plays a crucial role in amplifying risks. Online news is typically disseminated through users' social networks, and the efficacy of risk amplifications can be influenced by various characteristics of these networks. One such characteristic is social tie variety, which captures the overall pattern of relationships among individuals (Omar *et al.*, 2024) and measured as the diversity within users' social networks due to its significance in intra-network relationships and involvement in news verification behaviors (Torres *et al.*, 2018). As this diversity increases, individuals are exposed to diverse viewpoints from different backgrounds within their network, prompting them to critically evaluate the risks associated with disseminated news within that particular network (Torres *et al.*, 2018). The users with a higher diversity of social ties are more likely to perceive greater reputation losses when they share low-quality news. Furthermore, conflicting viewpoints may lead at least one piece of information to appear false to users, thereby increasing their perception of risk associated with news sharing. However, our understanding of the impact of social tie variety on the relationship between news quality and users' news-sharing behavior remains limited when incorporating users' risk perceptions. Therefore, we propose the third research question: *How does social tie variety moderate the effect of news quality on users' news-sharing behavior, mediated by risk perception and perceived believability?*

To address our aforementioned research questions, we employed a questionnaire to collect data and utilized the structural equation model to examine the relationships among constructs. This study contributes to both information system research and fake news studies by providing a comprehensive analysis of the mediating role of risk perception in influencing users' news-sharing behavior based on varying levels of news quality. Furthermore, this study employs the SARF framework to analyze the roles of fake news awareness and social tie variety as risk amplifications, while examining their moderating effects on the indirect effect of news quality on users' news-sharing behavior through risk perception and perceived believability.

The structure of this paper is as follows: Section 2 provides a comprehensive review of the relevant literature to identify the research gap and introduces the theoretical background. In Section 3, we propose a research model and formulate research hypotheses. Section 4 outlines the research methodology employed in this study. Subsequently, in Section 5, we present the data analysis and corresponding results obtained from our investigation. Finally, in Section 6, we engage in an extensive discussion and draw conclusions based on our findings.

2. Literature review and theoretical background

The section commences with a comprehensive literature review on the determinants influencing users' news-sharing behaviors, thereby identifying the research gap. Subsequently, we elaborate on the fundamental theory that underpins our investigation, namely the SARF, aiming to elucidate our research model.

2.1 Factors influencing users' news-sharing behaviors

The issue of news sharing has garnered significant attention due to the proliferation of fake news on the Internet. The rapid dissemination of fake news is widely recognized as a primary threat to societal well-being, as emphasized by the World Economic Forum (Wu *et al.*, 2022). Previous studies investigating factors influencing users' news-sharing behaviors can be broadly categorized into two groups: those focusing on news-related factors and those examining sharer-related factors.

The quality, type, and presentation format of news play critical roles in influencing users' news-sharing behavior. Previous studies have confirmed the impact of news quality on users' news-sharing behavior (Shin *et al.*, 2024), while also exploring the mediating effects of perceived believability (Kumar *et al.*, 2023) and perceived usefulness (Bhagat and Kim, 2023). News quality is conceptualized in terms of normative news values, including perceived factuality and perceived public importance, as well as social media values, such as perceived popularity (Shin *et al.*, 2024). It is also considered a moderating factor (Bhagat and Kim, 2023; Bi *et al.*, 2021; Thompson *et al.*, 2020) and mediating factor (Lee and Kim, 2023) in the relationship between user characteristics and users' news-sharing behavior. Furthermore, additional factors such as the moderating influence of rumor type on the association between personal involvement, trust, and intention to share have been examined (Chua and Banerjee, 2018). Specifically, dread-inducing online health rumors were found to elicit stronger share intention among medical professionals compared to wish-related rumors. It has been established that users' intention to share is directly influenced by perceived trustworthiness (Buchanan and Benson, 2019), with false information often propagated by unknown or low-rated sources (Kim *et al.*, 2019). Additionally, altering presentation formats emphasizing source credibility would impact users' news-sharing behavior through the mediating role of perceived believability (Kim and Dennis, 2019). Integrating intuitive icons in source evaluations to indicate the credibility of news sources can effectively mitigate the dissemination of false information (Kim *et al.*, 2023).

The proliferation of fake news can largely be attributed to user behavior, as individuals encounter a plethora of information through social media and unconsciously propagate it. Therefore, in addition to the factors associated with the news itself, scholars also focus on sharer-related factors, such as exposure to fact-check information, frequency of engagement in online discussions (Ho *et al.*, 2022), fake news awareness, social tie variety, and users' knowledge and attitudes. Specifically, users' knowledge and attitude towards the topic exerted a significant positive impact on their sharing behavior (Schaewitz *et al.*, 2020). Fake news awareness and social tie variety significantly enhance users' inclination to verify news (Torres *et al.*, 2018). Fake news awareness has also been considered a moderator and it has been observed that an increase in awareness weakens the relationship between user factors and online environmental factors in relation to users' news-sharing behavior (Omar *et al.*, 2024).

Table 1 provides a comprehensive overview of current research investigating various factors that influence users' news-sharing behavior. Previous studies have predominantly focused on the mediating roles of perceived believability and perceived usefulness in the relationship between news quality and users' news-sharing behavior, while neglecting the analysis of risk perception. Unraveling the influence of risk perceptions would enable a deeper understanding and intervention into users' intrinsic motivations for disseminating online news in an environment with abundant low-quality news. Additionally, although there has been partial examination of fake news awareness and social tie variety, it remains unclear whether these variables moderate the impact of news quality on risk perception and subsequent users' news-sharing behavior. Examining these two moderators would enhance our understanding of the risk amplification process and facilitate targeted intervention strategies to mitigate excessive sharing of fake news.

Table 1. A summary of influential factors impacting users' news-sharing behavior

Factors		The roles affecting user sharing behavior		Articles
News related factors	News quality	As IV	Mediated by perceived believability and perceived useful	Kumar et al. (2023) , Bhagat and Parrish (2018)
		As MO	Status seeking gratification is IV.	Thompson et al. (2020)
	Presentation format	As IV	Mediated by perceived believability	Kim and Dennis (2019)
	News source	As IV	n/a	Buchanan and Benson (2019) , Kim et al. (2019)
	Information type	As MO	Personal involvement is IV.	Chua and Banerjee (2018)
Sharer related factors	Fake news awareness	As IV	n/a	Torres et al. (2018)
		As MO	User factor and online environment factor are IVs	Omar et al. (2024)
	Social tie variety	As IV	n/a	Torres et al. (2018)
	Users' knowledge and attitude	As IV	n/a	Schawewitz et al. (2020)

Source(s): Authors' own work

2.2 Social amplification of risk framework

Considering the detrimental effects of low-quality online news, users are likely to perceive amplified risks and exercise caution not only towards such news but also towards all online news. Therefore, this study employed the SARF to analyze how risk perceptions regarding news quality influence users' news-sharing behaviors.

The SARF aims to incorporate pertinent factors that shape public reactions towards specific risks or risk events ([Beareth and Siegrist, 2022](#)). According to the SARF, it elucidates the intricate social processes underlying risk perception and response, as well as the interplay between risk signals and psychological, social, institutional, and cultural perspectives on risk perception and behavior ([Masuda and Garvin, 2006](#)). Risk amplification occurs in two stages: during the transmission of risk information and through social responses ([Kasperson et al., 1988](#)). The transmission stage involves individuals and social entities known as "amplification stations", such as opinion leaders, personal networks, and social media platforms ([Wirz et al., 2018](#)). This stage analyzes how risk information is communicated, framed, and interpreted by individuals and communities. Our focus lies in analyzing news quality's impact on users' perception of risks. Without the amplification of risks, users would solely perceive the risks associated with specific low-quality news sources. For instance, when encountering an email claiming a security issue with their account and requiring verification through a link click, users may only consider the risk involved in such clicking behavior. However, through the process of risk amplification, users might become concerned about potential virus-infected attachments in their emails and even opt to delete all messages from unfamiliar senders. Similarly, within the highly interconnected realm of the internet, users would experience an amplified perception of risks and exercise caution not only towards apparent low-quality news but also towards all online news.

The stage of social responses examines how the perception of amplified risk influences behavioral responses, resulting in secondary impacts or "ripples" ([Kasperson et al., 2022](#)), including emotional reactions, behavioral changes and societal consequences. This particular stage focuses on the influence of perceived risks on users' news-sharing behaviors mediated by their perception of believability. In our study, when users perceive a heightened risk

associated with online news sources, they are more likely to prioritize verifying the credibility of such sources and subsequently reduce their intentions to share online news.

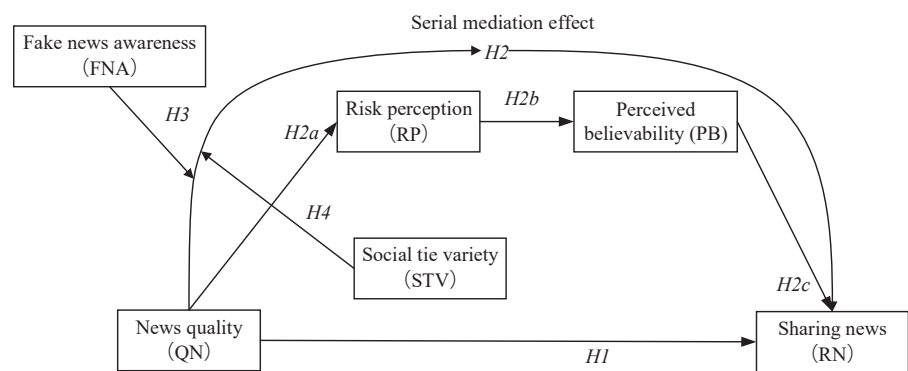
3. Research model and hypotheses

Building upon the SARF, this study investigates the influence of news quality on users’ news-sharing behavior through a sequential mediation process involving risk perception and perceived believability, while also exploring the moderating effects of fake news awareness and social tie variety. The research model is presented in Figure 1.

3.1 News quality and users’ news-sharing behavior

In an era marked by the prevalence of fake news, news quality becomes a concerning factor that impacts users’ news-sharing behavior. News quality incorporates the core components of accuracy, timeliness, and completeness, all of which contribute to the relevant and reliable of news (Nicolaou and McKnight, 2006). Moreover, high-quality information is devoid of errors and fulfills user’s needs (Gelle and Karhu, 2003). Building upon the Elaboration Likelihood Model, several studies have regarded news quality as a central processing cue (Bhagat and Kim, 2023) and have substantiated its direct impact on users’ attitudes and behaviors (Chen et al., 2024; Zhou et al., 2010). The provision of high-quality news offers pertinent, credible, timely, and precise information that enhances users’ confidence in making optimal decisions (Jiang et al., 2021). For instance, within the context of electronic information usage environment, news quality positively influences users’ adoption of information (Jiang et al., 2021; Zhou et al., 2023), purchase decisions (Fu et al., 2020) and intention to recommend it to others (Shin and Cheng, 2023). Similarly, in the domain of news dissemination, Cheng et al. (2019) affirmed that sharing quality-assured news exerts a positive impact on trust formation and intention for social shopping. Disseminating high-quality news not only provides recipients with valuable and timely information but also enhances senders’ status, reputation, and image (Zheng et al., 2013). However, when disseminating low-quality news items, spreaders may face numerous risks such as erosion of personal social status (Thompson et al., 2020) and reputational harm (Bhagat and Kim, 2023), thus making them more inclined towards sharing high-quality content rather than low-quality ones due to concerns about compromising their reputation through time-wasting activities. Therefore, we propose the following hypothesis:

H1. News quality is positively associated with users sharing behavior; indicating that when the perceived quality of a piece of news is sufficiently high, it will motivate users to share it.



Source(s): Authors’ own work

Figure 1. Research model

3.2 Risk perception and perceived believability as mediators

We employ the SARF to further investigate the impact of risk perception and perceived believability on the relationship between news quality and users' news-sharing behavior, wherein risk amplification manifests in two distinct stages: during the dissemination of risk information and through subsequent social responses (Kasperson *et al.*, 1988).

Risk perception refers to individuals' subjective expectations of potential losses associated with a specific risk event, such as news sharing activities in this study (Forsythe and Shi, 2003). It plays a crucial role in shaping the perspective on risk and subsequently influencing decision-making (Chen *et al.*, 2020; Yang *et al.*, 2015). High-quality news is characterized by its relevance, timeliness, accuracy, and comprehensiveness (Yi *et al.*, 2013), which helps reduce risks and uncertainties related to social status (Thompson *et al.*, 2020) and reputations (Bhagat and Kim, 2023) during news-sharing activities. A strong belief in the accuracy, currency, and relevance of the news increases its value when shared and consequently mitigates risk perception (Nicolaou and McKnight, 2006). Therefore, in an era marked by the prevalence of fake news, news quality is considered a signal of risk that influences users' perceptions of believability and their behaviors regarding news sharing. Therefore, we propose the following hypothesis:

H2a. News quality is negatively associated with users' risk perception.

Perceived believability reflects the extent to which users place trust in news content (Kukar-Kinney and Walters, 2003). The relationship between risk perception and perceived believability has been a subject of debate in prior research, with conflicting views on the direction of this relationship (Yi *et al.*, 2013). While some studies argue that perceived believability influences risk perception (Kim *et al.*, 2008; Nicolaou and McKnight, 2006), others assert that risk perception affects perceived believability (Koller, 1988; Nooteboom *et al.*, 1997; Ortega Egea and Román González, 2011). Mitchell (1999) provides a potential explanation for this apparent contradiction by emphasizing that risk perception is a prerequisite for trust to be operative, and one of the outcomes of building trust is a reduction in the perceived transactional or relational risk. According to this perspective, the impact of perceived believability on risk perception arises when users critically evaluate the construction of perceived believability and become conscious of potential risks during this process. However, since this study does not investigate the specific mechanisms underlying the formation of perceived believability, we do not consider the possibility that perceived believability directly influences risk perception.

Moreover, building upon the SARF, this study examines perceived believability as a user attitude during the process of how risk perception affects users' news-sharing behavior. We posit that risk perception drives users to evaluate news and their sharing behavior. Drawing from decision-making literature under risky conditions, we emphasize the pivotal role played by risk perception in shaping individuals' attitudes and response behaviors (Chen *et al.*, 2011). When users perceive potential risks, they engage in further evaluation of the source. For example, when consumers perceive risks associated with online shopping, they invest more time evaluating products and may refrain from future shopping activities (Forsythe and Shi, 2003). Similarly, when users perceive risks associated with email activities, they develop negative attitudes towards commercial emails and avoid reading them (Chen *et al.*, 2011). Likewise, when users perceive high levels of uncertainty and risks related to user sharing activities, they engage in identifying severity levels of these risks including verifying news credibility (Torres *et al.*, 2018). Therefore, this study considers risk perception as a precursor to perceived believability, and proposes the following hypothesis:

H2b. Users' risk perception is negatively associated with users' perceived believability.

Perceived believability is a prevalent psychological characteristic that detrimentally impacts user engagement (Jin and Atkinson, 2021). In the context of news sharing, believability holds significant importance as users tend to refrain from sharing news they find unconvincing

(Zhang and Cozma, 2022). Users only adopt news to aid decision-making if they perceive it as credible (Jiang *et al.*, 2021). If users do not believe the news to be true, their likelihood of sharing it diminishes (Kim and Dennis, 2019). Therefore, we propose the following hypothesis:

H2c. Users' perceived believability is positively associated with users' news-sharing behavior.

In summary, news quality serves as a risk indicator that directly triggers users' risk perceptions, which act as antecedents for assessing news believability (Koller, 1988) and subsequently influencing user intentions to share news. Based on this rationale, we propose the following hypothesis:

H2. News quality has a positive influence on sharing behavior through a serial mediating role of risk perception and perceived believability. The sequential pathway is defined as follows: news quality→risk perception→perceived believability→sharing news.

3.3 Fake news awareness as a moderator

Fake news awareness encompasses users' comprehension of the characteristics of fake news and their ability to discern them (Apuke *et al.*, 2022). Variations in individuals' awareness of fake news arise from their unique capacities and experiences (Torres *et al.*, 2018). Individuals lacking critical thinking skills and fake news awareness are more susceptible to fake news (Pennycook and Rand, 2019). When users possess heightened levels of fake news awareness, they adopt a more skeptical mindset towards online disseminated news (Majerczak and Strzelecki, 2022). The quality of news is determined by its accuracy and reliability, which translate into potential risks and uncertainties (Nicolaou and McKnight, 2006). In the context of low-quality news, individuals with enhanced fake news awareness tend to assign lower ratings to its accuracy and reliability compared to those with limited awareness, who may face challenges in assessing associated risks when disseminating such content (Torres *et al.*, 2018). Those with high levels of fake news awareness exhibit greater concern for reputational harm, leading them to perceive higher risks and uncertainties associated with user sharing activities compared to those with low levels of awareness (Apuke *et al.*, 2022). Consequently, individuals with high levels demonstrate significantly increased sensitivity towards declining quality when perceiving elevated dissemination risks.

Additionally, while fake news awareness moderates the impact of news quality on risk perception, it also influences subsequent perceived believability and leads to consequential behavioral changes. When users possess a high level of fake news awareness, they are more likely to perceive greater risks from low-quality news and have lower levels of news believability, resulting in a decreased likelihood to share such content. Therefore, we propose the following hypotheses:

H3. The indirect effect of news quality on users' news-sharing behavior through risk perception and perceived believability is negatively moderated by fake news awareness; this indirect effect is stronger for those with high levels of awareness and weaker for those with low levels.

3.4 Social tie variety as a moderator

Social media plays a pivotal role in amplifying social phenomena and exerting significant influence on individuals' risk perceptions by facilitating their exposure to risk information (Dai *et al.*, 2022). The advent of internet applications, such as social network sites, has empowered individuals to freely and expeditiously exchange information, establish networks based on shared interests, and enhance their social capital through the dissemination of information (Shen *et al.*, 2016).

Social tie variety, a structural dimension capturing the overall pattern of relationships among individuals (Omar *et al.*, 2024), is defined and measured as the diversity of users' social network (Torres *et al.*, 2018). An individual's online network encompasses family members, work colleagues, friends, and netizens sharing similar interests. As the diversity within these networks increases, there is a corresponding augmentation in the representation of diverse backgrounds within the network. This implies an enhanced exposure to divergent viewpoints (Torres *et al.*, 2018). According to SARE, contradictory risk messages amplify perceived risk and subsequently influence user behaviors. In other words, the proliferation of contrasting perspectives may prompt users to question news veracity disseminated within their networks (Majerczak and Strzelecki, 2022). Furthermore, conflicting viewpoints render news at least appear false or unreliable to users; thereby amplifying their perception regarding risks associated with sharing news. Consequently, compared with situations characterized by low social network diversity when users possess socially homogeneous ties only, their perceived risk pertaining to dissemination significantly escalates upon deterioration in news quality.

Furthermore, akin to the influence mechanism of fake news awareness, social tie variety not only moderates the impact of news quality on risk perception but also influences subsequent perceived believability and leads to consequential behavioral changes. When a user's social tie variety is high, they are more inclined to perceive greater risks from low-quality news and exhibit lower levels of news believability, resulting in a decreased likelihood to share such content. Therefore, we propose the following hypotheses:

- H4. The indirect effect of news quality on users' news-sharing behavior through risk perception and perceived believability is negatively moderated by social tie variety; this indirect effect is stronger for individuals with high social tie variety and weaker for those with low social tie variety.

4. Research methodology

4.1 Measurement development

The measurement items for all constructs were adapted from previous literature to suit the specific context of this study. Specifically, the measurement of news quality as an independent variable was adapted from Nicolaou and McKnight (2006). The measurements for fake news awareness and social tie variety as moderators were adapted from Torres *et al.* (2018). The measurements for perceived believability as an underlying mechanism were adapted from Kim *et al.* (2019), while the measurements for risk perception were adapted from Chen *et al.* (2011). Additionally, we measured users' news-sharing behavior by adapting the measurements for sharing news from Choi *et al.* (2017).

Previous studies have indicated that individuals with lower levels of education (Schaewitz *et al.*, 2020), younger age groups (Allcott and Gentzkow, 2017), and male gender (Filukova *et al.*, 2021) tend to actively participate in the dissemination of online news. To ensure the robustness of our findings, we included gender, age, and education as control variables in this study. Specifically, these variables were categorized as follows: Gender: 1 = Male; 0 = Female. Age: 1 = Below 20; 2 = Between 21–30; 3 = Between 31–40; and 4 = Above 40. Education: 1 = Junior middle school or below; 2 = Senior middle school; 3 = Undergraduate; 4 = Postgraduate.

Given that Credamo is a Chinese questionnaire platform, we recruited participants from China and administered the questionnaire in Chinese. Similar to Chen *et al.* (2023), we translated the original English measurements into Chinese and made slight modifications to ensure their appropriateness for our study. Prior to distributing the Chinese version of the questionnaire on Credamo, we conducted a pretest involving five respondents and made necessary adjustments to enhance its readability. Subsequently, during the writing process of this paper, we employed an English-Chinese-English translation procedure to render these

442

Table 2. Definitions and measurement items for constructs

Construct (Abbreviation)	Definition	Measurement items (1–7 Likert scale)	References
News quality (NQ)	Cognitive beliefs about the favorable or unfavorable characteristics of the currency, accuracy, completeness, relevance and reliability of news	NQ1: The news in my social network is complete NQ2: The news in my social network is clear and understandable NQ3: The news in my social network is well structured	Nicolaou and McKnight (2006)
Fake news awareness (FNA)	Users’ awareness of the existence of fake news in their social networks	FNA1: I am aware of the existence of fake news FNA2: I have heard about fake news FNA3: I am aware that some news in social network is fake FNA4: I know some of the news in social network is not true	Torres et al. (2018)
Social tie variety (STV)	The diversity of offline groups and contexts represented in one’s online social network	STV1: People I am connected to on my social network represent various activities I am involved in STV2: People I am connected to on my social network represent multiple stages of my life STV3: People I am connected to on my social network are diverse in how I met them	Torres et al. (2018)
Perceived believability (PB)	The extent to which news are regarded as true, real, and credible	PB1: The news in my network is believable to me PB2: The news in my social network is credible to me PB3: The source of news in my social network is credible PB4: The source of news in my social network is trustworthy	Kim and Dennis (2019)
Risk perception (RP)	Subjective expectations of potential losses associated with specific risk events (e.g. news sharing activities in this study)	RP1: My decision to share news in my social network is risky RP2: There is a high potential for loss involved by sharing news in my social network RP3: Sharing news in my social network will lead to considerable risks	Chen et al. (2011)
Sharing news (SN)	A fundamental aspect of news-related activities on social media, facilitating the dissemination of news to users’ social networks through the act of posting, forwarding, and sharing links	SN1: I often share news in my social media SN2: I often recommend news in my social media SN3: I always share news to my friends	Choi et al. (2017)
Source(s): Authors’ own work			

4.2 Data collection

Questionnaires were distributed and participants were recruited through Cremado, a professional Chinese survey platform with a sample size exceeding three million, comparable to Amazon Mechanical Turk. Previous research has demonstrated that the results obtained from the Cremado data platform survey exhibit no statistically significant differences when compared to face-to-face and email questionnaires, thereby establishing the reliability of data collected from Cremado (Chen *et al.*, 2023). Notably, several recent studies utilizing data from [Credamo.com](https://www.credamo.com) have been published in prestigious journals such as *Journal of Consumer Research*, *Journal of Marketing Research*, and *Production and Operations Management*.

All questions were set as mandatory, and the Credamo platform exclusively returned completed questionnaires, ensuring no missing values in our data. We finally collected 399 valid questionnaires, and the sample profiles of the responses are presented in Table 3. Among the 399 applicable participants, the majority (63.4%) were female. More than half of them (56.1%) fell within the age range of 21–30, while a substantial majority (82.7%) had achieved high levels of education through university attendance. Notably, this demographic comprised an active cohort on social networks.

5. Data analysis and results

We employed a two-step approach to examine the proposed model. Firstly, we conducted an exploratory factor analysis (EFA) and confirmatory factor analysis (CFA) using MPLUS to evaluate our measurement model based on multiple fit criteria, reliability, and validity. Subsequently, we assessed the structural model and tested the hypothesized relationships.

5.1 Measurement model

5.1.1 Unidimensionality and reliability. To ensure the reliability of our constructs, we employed the method proposed by Narasimhan and Jayaram (1998). Firstly, EFA was conducted to confirm the unidimensionality of our constructs. We utilized the maximum likelihood method and Geomin Rotation for EFA (Table 4).

The results revealed that all items exhibited higher loadings on their respective measured constructs and lower loadings on unmeasured constructs, indicating the unidimensionality of the constructs. Secondly, Cronbach's alpha and composite reliability (CR) were computed for each construct to assess construct reliability. As presented in Table 5, all Cronbach's alpha and composite reliability values exceeded the critical threshold of 0.70, demonstrating a high level of internal consistency for each construct. Additionally, the corrected item-total correlation

Table 3. Respondents' demographics

		Number	Percent
Gender	Male	146	36.6%
	Female	253	63.4%
Age	≤20	12	3%
	21–30	224	56.1%
	31–40	137	34.4%
	≥41	26	6.5%
Education	Junior high school or below	1	0.3%
	Senior high school	10	2.5%
	Undergraduate	330	82.7%
	Postgraduate	58	14.5%

Source(s): Authors' own work

Table 4. Geomin rotated loadings for constructs in EFA

Construct	Item	NQ	FNA	STV	PB	RP	SN
News quality	QN1	0.60	−0.04	0.06	0.31	−0.002	0.07
	QN2	0.82	−0.01	−0.03	−0.01	0.01	0.05
	QN3	0.60	−0.01	0.01	0.2	−0.13	−0.05
Fake news awareness	FNA1	−0.04	0.82	−0.05	−0.13	−0.02	0.01
	FNA2	0.03	0.87	0.03	0.01	0.05	−0.05
	FNA3	−0.06	0.91	0.02	−0.02	−0.04	0.02
	FNA4	0.02	0.88	0.03	0.04	0.07	−0.02
Social tie variety	STV1	0.08	0.07	0.72	−0.05	−0.04	−0.002
	STV2	0.02	0.04	0.60	0.12	0.14	0.03
	STV3	−0.1	−0.09	0.72	−0.01	−0.02	−0.01
Perceived believability	PB1	−0.01	−0.09	0.04	0.84	0.03	0.02
	PB2	−0.03	0.04	−0.08	0.93	−0.01	0.01
	PB3	0.05	−0.08	0.07	0.70	−0.08	0.06
	PB4	0.10	0.04	−0.05	0.77	−0.10	0.01
Risk perception	RP1	0.02	0.03	−0.05	−0.11	0.84	0.06
	RP2	−0.05	−0.04	0.02	−0.02	0.83	0.01
	RP3	−0.02	0.02	0.02	0.01	0.83	0.06
Sharing news	RE1	0.04	0.01	0.03	−0.04	−0.04	0.90
	RE2	0.001	0.01	−0.02	0.06	−0.03	0.87
	RE3	−0.01	−0.05	0.003	0.06	0.04	0.87

Source(s): Authors' own work

(CITC) for each construct significantly surpassed the critical value of 0.3. Consequently, all examined constructs in this study exhibited satisfactory levels of reliability.

5.1.1.2 Construct validity. Construct validity refers to the degree to which a measurement accurately captures a theoretical concept, encompassing content, convergent, and discriminant validity. To ensure the content validity of our scale, we conducted a comprehensive review of relevant literature and adapted established measures for the independent variable, moderators, dependent variables, and mediators. The measurement model consists of 6 factors with their corresponding 20 items presented in Table 2.

To establish convergent and discriminant validity, we conducted CFA using multiple fit criteria including the root mean square error of approximation (RMSEA), comparative fit index (CFI), Tucker-Lewis index (TLI), and standardized root mean square residual (SRMR). Based on previous literature, acceptable levels of fit were defined as follows: $RMSEA \leq 0.06$; $CFI \geq 0.95$; $TLI \geq 0.95$; $SRMR \leq 0.08$. Our data met these criteria with satisfactory results for all four indices: $\chi^2(155) = 344.54$, $RMSEA = 0.055 \leq 0.06$, $CFI = 0.972 \geq 0.95$, $TLI = 0.965 \geq 0.95$, $SRMR = 0.032 \leq 0.08$, indicating good model fit reliability and validity for subsequent tests of our proposed hypotheses. Convergent validity was assessed by computing average variance extracted (AVE) which showed that while the AVE value for society tie variety was marginal at 45%, all other constructs had values higher than 50% (Bagozzi and Yi, 1988; Fornell and Larcker, 1981). Additionally, factor loadings above 0.50 indicated strong convergent validity across all items as shown in Table 5. Discriminant validity was established by ensuring that the square root of each construct's AVE value exceeded its correlation coefficients with other constructs as shown in Table 6 (Fornell and Larcker, 1981). These results confirmed acceptable discriminant validity.

Furthermore, preliminary hypothesis testing revealed positive relationships between news quality and users' news-sharing behavior ($r = 0.63, p < 0.001$); negative relationships between news quality and risk perception ($r = -0.62, p < 0.001$); negative relationships between risk perception and perceived believability ($r = -0.73, p < 0.001$); and positive relationships between perceived believability and users' news-sharing behavior ($r = 0.71, p < 0.001$).

Table 5. Reliability analysis

Construct	Standardized loading	Cronbach's alpha	C.R.	AVE	CITC
News quality		0.83	0.84	0.63	0.693–0.704
NQ1	0.84				
NQ2	0.75				
NQ3	0.79				
Fake news awareness		0.938	0.94	0.79	0.835–0.877
FNA1	0.87				
FNA2	0.88				
FNA3	0.92				
FNA4	0.89				
Social tie variety		0.71	0.71	0.45	0.476–0.544
STV1	0.74				
STV2	0.60				
STV3	0.67				
Perceived believability		0.935	0.94	0.79	0.840–0.854
PB1	0.87				
PB2	0.88				
PB3	0.90				
PB4	0.90				
Risk perception		0.925	0.93	0.81	0.843–0.86
RP1	0.89				
RP2	0.89				
RP3	0.92				
Sharing news		0.939	0.94	0.84	0.867–0.879
SN1	0.91				
SN2	0.93				
SN3	0.91				

Note(s): C.R. = composite reliability; AVE = average variance extracted; CITC = corrected item-total correlation

Source(s): Authors' own work

Table 6. Properties of measurement scales

Construct	ME	SD	Correction					
			1	2	3	4	5	6
1.NQ	5.51	1.06	0.79					
2.FNA	5.21	1.35	−0.35***	0.89				
3.STV	6.01	0.74	0.21***	0.01	0.67			
4.RP	3.03	1.39	−0.62***	0.41***	−0.04	0.90		
5.PB	5.13	1.14	0.74***	−0.45***	0.12*	−0.73***	0.89	
7.SN	4.85	1.63	0.63***	−0.41***	0.15***	−0.57***	0.71***	0.92

Note(s): 1. ME = mean; SD = standard deviation

2. The diagonal value in bold print is the square roots of the AVEs

3. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Source(s): Authors' own work

The preliminary results of the correlation test have provided support for our hypotheses and established a foundation for subsequent hypothesis testing.

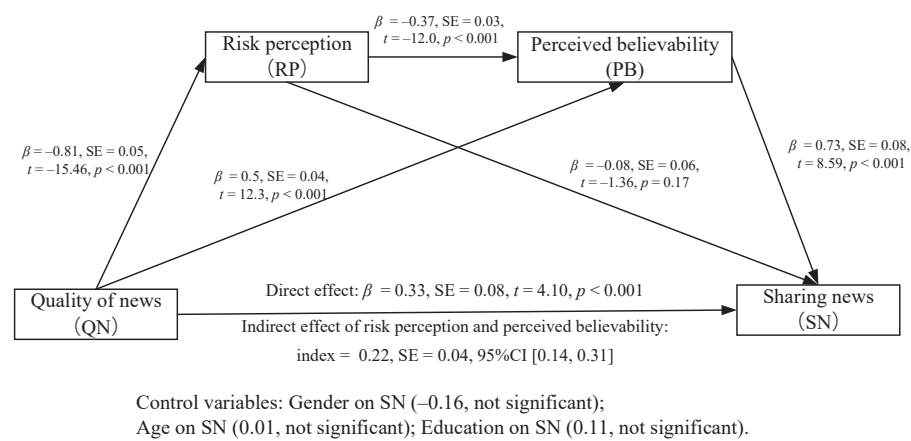
5.1.3 Common method bias. The potential occurrence of common method bias may arise due to the utilization of self-reported data (Podsakoff and Organ, 1986). To address this concern, we initially conducted Harman's one-factor test. The results revealed six distinct factors, with the largest factor explaining 46.69% of the covariance, which is below the

threshold of 50% (Howard and Henderson, 2023). Additionally, a common method factor analysis was performed to further assess possible common method bias (Podsakoff *et al.*, 2003). The findings indicated that the loadings of substantive factors were statistically significant at the 5% level ($p < 0.05$), while those of the method factor were not significant. Initial CFA yielded CFI and TLI values of 0.972 and 0.965, respectively, along with RMSEA and SRMR values were 0.055 and 0.032, respectively. When incorporating the common method factor in CFA, revised results showed improved CFI and TLI values (0.977 and 0.968 respectively), as well as reduced RMSEA and SRMR values (0.053 and 0.023 respectively). The changes in CFI/TLI are within an acceptable range (<0.1 increase), and the alterations in RMSEA/SRMR are also minimal (<0.05 decrease). Therefore, we can conclude that common method bias is not a significant concern in this study.

5.2 Hypothesis testing

5.2.1 Serial mediation model. To empirically examine our theoretical model, we conducted a serial mediation analysis using PROCESS Model 6 (Hayes, 2017), with news quality as the independent variable and sharing news as the dependent variable. Figure 2 illustrates the path analysis. The findings revealed that the proposed model accounted for 54% of the variance ($R^2 = 0.54$, $p < 0.001$) in sharing news. Specifically, regression results confirmed a significant positive effect of news quality on users' sharing behavior ($\beta = 0.33$, $p < 0.001$), supporting H1. Additionally, the findings also substantiated a significant inverse impact of news quality on risk perception ($\beta = -0.81$, $p < 0.001$), a noteworthy negative influence of risk perception on perceived believability ($\beta = -0.37$, $p < 0.001$), and a substantial positive effect of perceived believability on users' sharing behavior ($\beta = 0.73$, $p < 0.001$), thereby supporting H2a, H2b and H2c.

Consistent with our Hypothesis 2, the serial mediation model demonstrated significant paths from news quality to risk perception, perceived believability, and sharing news. More specifically, there was an indirect effect of risk perception on users' sharing behavior (indirect effect = 0.07, SE = 0.06, 95%CI $[-0.05, 0.18]$). Additionally, perceived believability had an indirect effect on users' sharing behavior (indirect effect = 0.36, SE = 0.05, 95%CI $[0.26, 0.47]$). Furthermore, the combined indirect effects of risk perception and perceived believability on sharing news were found to be significant (indirect effect = 0.22,



Source(s): Authors' own work

Figure 2. The serial mediation role of RP and PB in the relationship between QN and SN

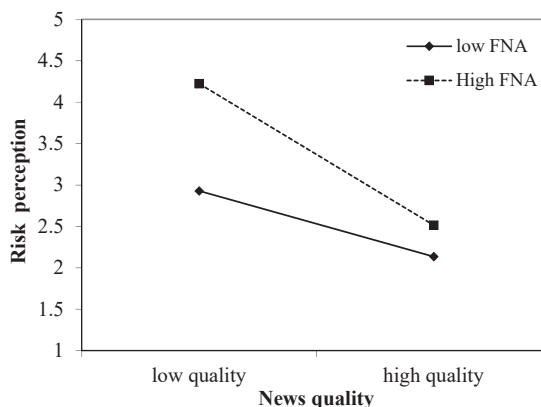
SE = 0.04, 95%CI [0.14, 0.31]), thus validating the serial mediation between risk perception and perceived believability. Hypothesis 2 is supported.

To further validate our theoretical framework that posits how news quality influences users' sharing behavior through sequential mediating roles of risk perception and perceived believability, we adjusted these two mediators in a sequential order. The results indicated that this adjusted chain mediating effect was not statistically significant ($\beta = 0.05$, 95%CI [-0.04, 0.13]), thereby confirming appropriateness of assuming a serial mediation model in this study.

5.2.2 Moderating effect of fake news awareness. To test [Hypothesis 3](#), we first conducted a moderation analysis using PROCESS Model 1 ([Hayes, 2017](#)), with news quality as the independent variable, risk perception as the dependent variable, and fake news awareness as the moderator. Prior to analysis, both fake news awareness and news quality were mean centered. The results revealed a significant interaction effect between news quality and fake news awareness ($\beta = -0.16$, $p = 0.002$), indicating that fake news awareness significantly moderates the relationship between news quality and risk perception. [Figure 3](#) illustrates the moderating effect of fake news awareness, demonstrating that high levels of fake news awareness led to a significant decrease in their risk perception associated with news quality, whereas low levels of fake new awareness weaken the negative relationship between news quality and risk perception.

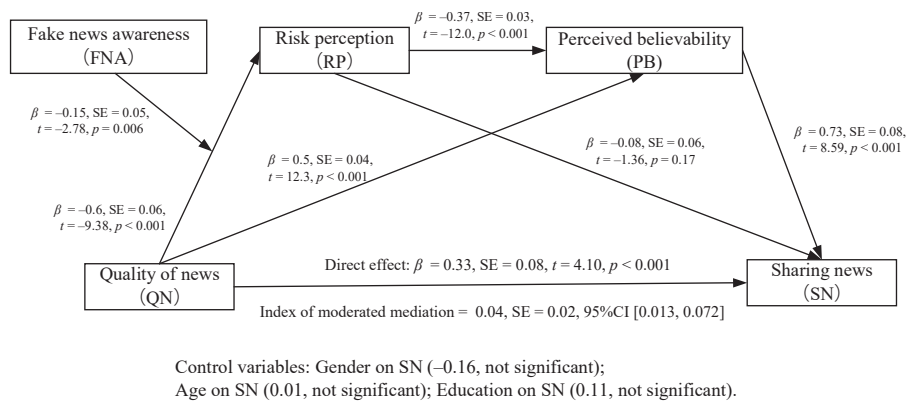
Furthermore, we conducted a serial moderated mediation analysis using PROCESS Model 83 ([Hayes, 2017](#)), with news quality as the independent variable, sharing news as the dependent variable, and fake news awareness as the moderator. Risk perception and perceived believability were included as mediators. Prior to analysis, both fake news awareness and news quality were mean centered. [Figure 4](#) illustrates the path analysis. Consistent with our assumption, bootstrapping with 5,000 samples revealed a significant moderated mediation effect of fake news awareness on the relationship between news quality and sharing news (index = 0.04, SE = 0.02, 95%CI [0.013, 0.072]). The robustness of the main effect is compromised from -0.81 to -0.59 when considering the moderating effect of fake news awareness, as it becomes contingent upon varying levels.

Specifically, as presented in [Table 7](#), when users exhibit high awareness of fake news, the serial mediation effect of risk perception and perceived believability on news quality and sharing news is statistically significant ($\beta = 0.21$, 95%CI [0.14, 0.30]). Similarly, when users have low awareness of fake news, the serial mediation effect of risk perception and perceived



Source(s): Authors' own work

Figure 3. The moderating effect of FNA in the relationship between QN and RP



Source(s): Authors' own work

Figure 4. Results of moderated mediation analysis of FNA in the relationship between QN and SN

Table 7. Comparison of serial mediation effects in different levels

Different level of FNA	Effect	SE	95%CI LICI	ULCI
High level of FNA	0.21	0.04	0.14	0.30
Low level of FNA	0.11	0.04	0.04	0.19
Diff (high-low)	0.11	0.04	0.03	0.19

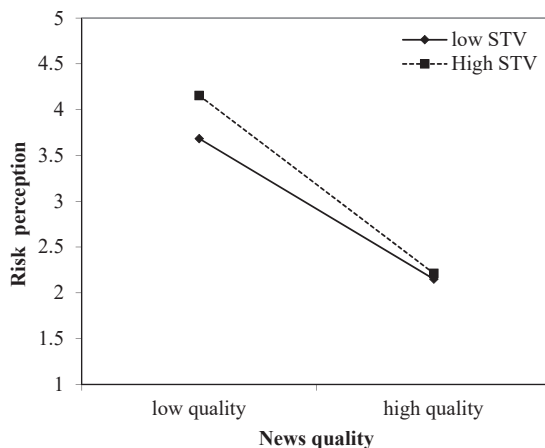
Note(s): High level refers to values that are one standard deviation above the mean; low level refers to values that are one standard deviation below the mean

Source(s): Authors' own work

believability on news quality and sharing news remains valid ($\beta = 0.11$, 95%CI [0.04, 0.19]). Notably, there are notable disparities in the serial mediation effect across different levels of fake news awareness ($\beta_{(\text{high-low})} = 0.11$, 95%CI [0.03, 0.19]). Thus, supporting [Hypothesis 3](#) that posits fake news awareness moderates the serial mediation effect of risk perception and perceived believability on news quality and sharing behavior.

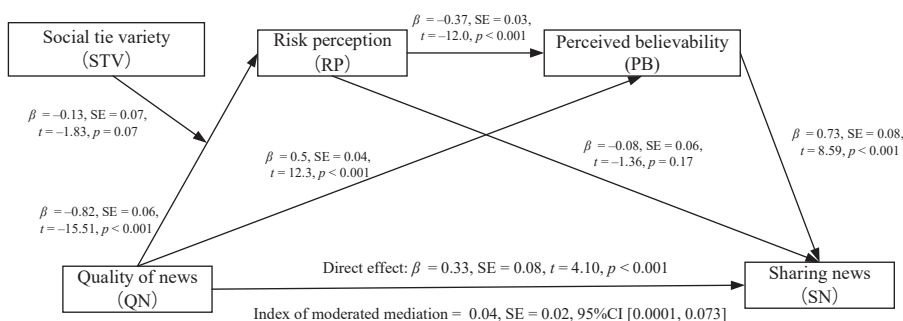
5.2.3 Moderating effect of social tie variety. To test [Hypothesis 4](#), we first conducted a moderation analysis using PROCESS Model 1 ([Hayes, 2017](#)), with news quality as the independent variable, risk perception as the dependent variable, and social tie variety as the moderator. Prior to analysis, both social tie variety and news quality were mean centered. The results indicated that there was a significant interaction effect between news quality and social tie variety ($\beta = -0.13$, $p = 0.05$), suggesting that social tie variety can moderate the relationship between news quality and risk perception. [Figure 5](#) illustrates this moderating effect of social tie variety: when users had high levels of social tie variety, their risk perception decreased significantly in response to higher-quality news; conversely, when users had low levels of social tie variety, the negative association between news quality and risk perception weakened.

Furthermore, we conducted a serial moderated mediation analysis using PROCESS Model 83 ([Hayes, 2017](#)), with news quality as the independent variable, sharing news as the dependent variable, social tie variety as the moderator, and risk perception and perceived believability as mediators. Prior to analysis, both social tie variety and news quality were mean centered. [Figure 6](#) illustrates the path analysis. Consistent with our assumption, bootstrapping



Source(s): Authors' own work

Figure 5. The moderating effect of STV in the relationship between QN and RP



Source(s): Authors' own work

Figure 6. Results of moderated mediation analysis of STV in the relationship between QN and SN

with 5,000 samples revealed a significant moderated mediation effect of social tie variety on the association between news quality and sharing news (index = 0.04 , SE = 0.02 , 95%CI [0.001 , 0.073]).

Specifically, as presented in Table 8, when a user's social tie variety is high, the serial mediation effect of risk perception and perceived believability on news quality and sharing news is significant ($\beta = 0.25$, 95%CI [0.16 , 0.35]). Similarly, when the variety of a user's social ties is low, the serial mediation effect of risk perception and perceived believability on news quality and sharing news remains statistically significant ($\beta = 0.20$, 95%CI [0.12 , 0.28]). Notably, there are notable differences observed in the serial mediation effect across different levels of social tie variety ($\beta_{\text{high-low}} = 0.05$, 95%CI [0.001 , 0.11]). Thus supporting Hypothesis 4 that posits social tie variety moderates the serial mediation effect of risk perception and perceived believability on news quality and sharing news.

Table 8. Comparison of serial mediation effects in different level

Different level of STV	Effect	SE	95%CI LICI	ULCI
High level of STV	0.25	0.05	0.16	0.35
Low level of STV	0.20	0.04	0.12	0.28
Diff (high-low)	0.05	0.03	0.0001	0.11

Note(s): High level refers to values that are one standard deviation above the mean; low level refers to values that are one standard deviation below the mean

Source(s): Authors' own work

6. Discussion and conclusion

Building upon the SARF, this study employs a questionnaire methodology to conduct moderating and mediating analyses in order to investigate the roles of risk perception, perceived believability, fake news awareness, and social tie variety in shaping the impact of news quality on users' news-sharing behavior. Our findings strongly support the theoretical model proposed in this study. The results of hypothesis testing are summarized comprehensively in [Table 9](#).

We have observed a positive correlation between news quality and the act of sharing news, which aligns with the assertion made by [Zheng et al. \(2013\)](#) that sharing high-quality news not only imparts recipients with valuable and timely information but also enhances the sender's status, reputation, and image.

To elucidate the underlying mechanism behind the well-established phenomenon of users' heightened inclination to share high-quality news, we present three novel findings. Firstly, the relationship between news quality and news sharing is sequentially mediated by risk perception and perceived believability. News quality serves as a risk indicator that directly triggers users' risk perceptions, which act as antecedents for assessing news believability ([Koller, 1988](#)) and subsequently influencing user intentions to share news. This differs from previous literature that has demonstrated the mediating role of perceived believability in the association between news quality and users' news-sharing behavior but lacks an analysis of risk perception ([Kumar et al., 2023](#)). Secondly, the moderating role of fake news awareness was validated, indicating that individuals with a heightened level of fake news awareness demonstrate increased concern regarding perceived risks associated with user sharing activities in comparison to those with lower levels of awareness. This finding diverges from prior research that has primarily examined how fake news awareness moderates the relationship between user factors or online environmental factors and users' news-sharing

Table 9. Summary of hypothesis testing

Hypotheses	Results
H1: News quality is positively associated with users' sharing behavior	Supported
H2a: News quality is negatively associated with users' risk perception	Supported
H2b: Users' risk perception is negatively associated with users' perceived believability	Supported
H2c: Users' risk perception is negatively associated with users' perceived believability	Supported
H2: News quality has a positive influence on sharing behavior through a serial mediating role of risk perception and perceived believability	Supported
H3: The indirect effect of news quality on user sharing behavior through risk perception and perceived believability is negatively moderated by fake news awareness	Supported
H4: The indirect effect of news quality on user sharing behavior through risk perception and perceived believability is negatively moderated by social tie variety	Supported

Source(s): Authors' own work

behavior (Omar *et al.*, 2024), without delving into the interaction effects between news quality and fake news awareness. Thirdly, the moderating role of social tie variety was substantiated, indicating that individuals with a higher diversity of social ties intensify their risk perception associated with news sharing and exhibit skepticism towards news veracity. This finding deviates from previous research primarily focused on the impact of social tie variety on news verification behaviors while overlooking its influence on users' news-sharing behavior (Torres *et al.*, 2018).

6.1 Theoretical implications

This study contributes to the existing literature on users' news-sharing behaviors and SARF by making two theoretical advancements. Firstly, this study investigates the influence of risk perception on shaping the relationship between news quality, perceived believability, and users' news-sharing behavior. Previous literature has demonstrated the direct impact of news quality on users' news-sharing behavior while also exploring the mediating role of perceived believability (Kumar *et al.*, 2023). However, it has overlooked the significance of examining risk perception. Unraveling the influence of risk perceptions would enable a deeper understanding and intervention into users' intrinsic motivations for disseminating online news in an environment with abundant low-quality news. Building upon the SARF, this study asserts a serial mediating role of risk perception and perceived believability between news quality and users' news-sharing behavior, while refuting the possibility that perceived believability precedes risk perception in the context of news-sharing activities.

Secondly, our analysis on the moderating effects of fake news awareness and social tie variety contributes to a deeper understanding of users' news-sharing behavior. Previous studies have primarily focused on how fake news awareness moderates the relationship between user factors or online environmental factors and users' news-sharing behavior (Omar *et al.*, 2024), without exploring the interaction effects between news quality and fake news awareness. Similarly, prior research has mainly examined the impact of social tie variety on news verification behaviors while neglecting its influence on users' news-sharing behavior (Torres *et al.*, 2018). Incorporating considerations of both fake news awareness and social tie variety is crucial for comprehending users' engagement in sharing news, as users' levels of fake news awareness remain relatively low and significantly vary despite the prevalence of fake news within their social networks (Wu *et al.*, 2022).

6.2 Practical implications

The advent of internet applications, such as social networking sites, has empowered individuals to actively and rapidly engage in sharing encountered information, thereby exacerbating the proliferation of fake news. Despite concerted efforts by public authorities, governmental bodies, and academic institutions to curtail the dissemination, satisfactory outcomes have yet to be achieved. The findings from this study hold significant implications for practitioners aiming to impede the propagation of fake news.

Firstly, the results from this study provide evidence for the positive impact of news quality on sharing behavior by establishing a sequential mediation model involving risk perception and perceived believability. Prior literature has advised key stakeholders including governmental organizations and mainstream media outlets to prioritize high-quality news content given its greater likelihood of being shared by users who value reliability (Bhagat and Parrish, 2018; Kumar *et al.*, 2023). Furthermore, it is recommended that social media platforms proactively participate in fact-checking endeavors due to the influential role played by perceived believability in shaping users' news-sharing behavior (Kumar *et al.*, 2023). This research contributes to existing knowledge by confirming the significance of incorporating risk perception while proposing social media initiatives aimed at fostering users' sense of societal responsibility and awareness regarding the potential risks associated with spreading fake news.

Secondly, our study demonstrates that when users possess a heightened awareness of fake news, their perceived risk of dissemination significantly increases in the face of declining news quality. Therefore, practitioners should identify individuals with low levels of fake news awareness and prioritize efforts to enhance their understanding in order to mitigate the proliferation of fake news. Consequently, governments can augment the number of informative lectures on the perils associated with sharing fake news to cultivate a broader societal consciousness among social network users, particularly targeting those who exhibit limited fake news awareness, such as young individuals lacking comprehensive knowledge (Wu *et al.*, 2022). Additionally, we can explore various educational settings including schools and workplaces to implement interactive games aimed at enhancing media literacy and fostering greater awareness regarding fake news (Roozenbeek and van der Linden, 2019). Given the limited discernment of users with low fake news awareness, who often fail to recognize the substantial risks associated with low-quality news, we propose that training programs aimed at enhancing fake news awareness should incorporate a greater number of risk statements pertaining to low-quality news.

Thirdly, our findings indicate that users with diverse social networks are more inclined to perceive elevated risks associated with the dissemination of low-quality news. Consequently, it is imperative for practitioners to prioritize individuals with limited social ties and provide them with enhanced training opportunities to comprehend the prevalence of fake news and the potential risks linked to its propagation.

6.3 Limitation and future research

This study also presents certain limitations, which provide opportunities for future research. Firstly, due to the utilization of survey methodology in this study, we did not establish a direct link between our constructs and specific news items. For example, news quality was operationalized by soliciting participants' evaluations regarding the quality of news within their social network, while sharing news was operationalized by assessing their frequency of news sharing on social media platforms. Employing an experimental design where participants are exposed to specific news items and capturing their responses would have facilitated drawing reasonably robust causal inferences. Secondly, the generalizability of this study may be affected due to its data collection in China. It is important to consider that culture plays a significant role in the dissemination of fake news. For instance, citizens in Saudi Arabia exhibit a relatively high awareness of COVID-19 and are hesitant to share relevant medical information. It is well-known that Western and Eastern cultures differ in terms of people's information processing, which can lead to variations (Wu *et al.*, 2022). Thirdly, the investigated sample predominantly comprised of highly educated young respondents, who constitute a substantial proportion of the online interaction population. However, future research could enhance the robustness of findings by expanding the sample size to encompass a more diverse range of participants. Fourthly, our research model fails to consider an additional aspect of social ties, namely social tie strength. It is important to note that users with strong or weak social ties may exhibit different preferences and engage in distinct news sharing behavior. Therefore, future research should explore the impact of social tie strength in order to provide deeper insights into news sharing dynamics.

References

- Allcott, H. and Gentzkow, M. (2017), "Social media and fake news in the 2016 election", *The Journal of Economic Perspectives*, Vol. 31 No. 2, pp. 211-236, doi: [10.1257/jep.31.2.211](https://doi.org/10.1257/jep.31.2.211).
- Apuke, O.D., Omar, B. and Tunca, E.A. (2022), "Effect of fake news awareness as an intervention strategy for motivating news verification behaviour among social media users in Nigeria: a quasi-experimental research", *Journal of Asian and African Studies*, Vol. 58 No. 6, pp. 888-903, doi: [10.1177/00219096221079320](https://doi.org/10.1177/00219096221079320).

- Bagozzi, R.P. and Yi, Y. (1988), "On the evaluation of structural equation models", *Journal of the Academy of Marketing Science*, Vol. 16 No. 1, pp. 74-94, doi: [10.1177/009207038801600107](https://doi.org/10.1177/009207038801600107).
- Bearth, A. and Siegrist, M. (2022), "The social amplification of risk framework: a normative perspective on trust?", *Risk Analysis*, Vol. 42 No. 7, pp. 1381-1392, doi: [10.1111/risa.13757](https://doi.org/10.1111/risa.13757).
- Bhagat, S. and Kim, D.J. (2023), "Examining users' news sharing behaviour on social media: role of perception of online civic engagement and dual social influences", *Behaviour and Information Technology*, Vol. 42 No. 8, pp. 1194-1215, doi: [10.1080/0144929x.2022.2066019](https://doi.org/10.1080/0144929x.2022.2066019).
- Bhagat, S. and Parrish, J. (2018), "Examining the role of information quality, personal/social relevance, current events expertise in news sharing on social media", *24th Americas Conference on Information Systems 2018: Digital Disruption, AMCIS 2018*, Association for Information Systems.
- Bi, N.C., Lu, Y., Ha, L. and Chen, P. (2021), "Attitude change toward the Chinese during the US-China trade conflict: examining the roles of social media news sharing and perceived news feed performance", *Online Information Review*, Vol. 45 No. 3, pp. 599-613, doi: [10.1108/oir-05-2020-0178](https://doi.org/10.1108/oir-05-2020-0178).
- Buchanan, T. and Benson, V. (2019), "Spreading disinformation on Facebook: do trust in message source, risk propensity, or personality affect the organic reach of "fake news"?", *Social Media + Society*, Vol. 5 No. 4, pp. 1-9, doi: [10.1177/2056305119888654](https://doi.org/10.1177/2056305119888654).
- Chen, R., Wang, J., Herath, T. and Rao, H.R. (2011), "An investigation of email processing from a risky decision making perspective", *Decision Support Systems*, Vol. 52 No. 1, pp. 73-81, doi: [10.1016/j.dss.2011.05.005](https://doi.org/10.1016/j.dss.2011.05.005).
- Chen, J.V., Biamukda, S. and Tran, S.T.T. (2020), "Service providers' intention to continue sharing: the moderating role of two-way review system", *Industrial Management and Data Systems*, Vol. 120 No. 8, pp. 1543-1564, doi: [10.1108/imds-08-2019-0436](https://doi.org/10.1108/imds-08-2019-0436).
- Chen, L., Wu, P., Dou, Y. and Wu, Y. (2023), "Investigating senders' switching intention to smart lockers: an extension of push-pull-mooring model", *Journal of Retailing and Consumer Services*, Vol. 74, 103414, doi: [10.1016/j.jretconser.2023.103414](https://doi.org/10.1016/j.jretconser.2023.103414).
- Chen, Q., Jin, J. and Yan, X. (2024), "Impact of online physician service quality on patients' adoption behavior across different stages: an elaboration likelihood perspective", *Decision Support Systems*, Vol. 176, 114048, doi: [10.1016/j.dss.2023.114048](https://doi.org/10.1016/j.dss.2023.114048).
- Cheng, X., Gu, Y. and Shen, J. (2019), "An integrated view of particularized trust in social commerce: an empirical investigation", *International Journal of Information Management*, Vol. 45, pp. 1-12, doi: [10.1016/j.ijinfomgt.2018.10.014](https://doi.org/10.1016/j.ijinfomgt.2018.10.014).
- Choi, J., Lee, J.K. and Metzgar, E.T. (2017), "Investigating effects of social media news sharing on the relationship between network heterogeneity and political participation", *Computers in Human Behavior*, Vol. 75, pp. 25-31, doi: [10.1016/j.chb.2017.05.003](https://doi.org/10.1016/j.chb.2017.05.003).
- Chua, A.Y.K. and Banerjee, S. (2018), "Intentions to trust and share online health rumors: an experiment with medical professionals", *Computers in Human Behavior*, Vol. 87, pp. 1-9, doi: [10.1016/j.chb.2018.05.021](https://doi.org/10.1016/j.chb.2018.05.021).
- Dai, Y., Huang, Y.-H.C., Jia, W. and Cai, Q. (2022), "The paradoxical effects of institutional trust on risk perception and risk management in the Covid-19 pandemic: evidence from three societies", *Journal of Risk Research*, Vol. 25 Nos 11-12, pp. 1337-1355, doi: [10.1080/13669877.2022.2108122](https://doi.org/10.1080/13669877.2022.2108122).
- Dennis, A.R., Moravec, P.L. and Kim, A. (2023), "Search and verify: misinformation and source evaluations in Internet search results", *Decision Support Systems*, Vol. 171, 113976, doi: [10.1016/j.dss.2023.113976](https://doi.org/10.1016/j.dss.2023.113976).
- Filkukova, P., Ayton, P., Rand, K. and Langguth, J. (2021), "What should I trust? Individual differences in attitudes to conflicting information and misinformation on COVID-19", *Frontiers in Psychology*, Vol. 12, 588478, doi: [10.3389/fpsyg.2021.588478](https://doi.org/10.3389/fpsyg.2021.588478).
- Fornell, C. and Larcker, D.F. (1981), "Evaluating structural equation models with unobservable variables and measurement error", *Journal of Marketing Research*, Vol. 18 No. 1, pp. 39-50, doi: [10.2307/3151312](https://doi.org/10.2307/3151312).

- Forsythe, S.M. and Shi, B. (2003), "Consumer patronage and risk perceptions in Internet shopping", *Journal of Business Research*, Vol. 56 No. 11, pp. 867-875, doi: [10.1016/s0148-2963\(01\)00273-9](https://doi.org/10.1016/s0148-2963(01)00273-9).
- Fu, J.-R., Lu, I.W., Chen, J.H.F. and Farn, C.-K. (2020), "Investigating consumers' online social shopping intention: an information processing perspective", *International Journal of Information Management*, Vol. 54, 102189, doi: [10.1016/j.ijinfomgt.2020.102189](https://doi.org/10.1016/j.ijinfomgt.2020.102189).
- Gelle, E. and Karhu, K. (2003), "Information quality for strategic technology planning", *Industrial Management and Data Systems*, Vol. 103 No. 8, pp. 633-643, doi: [10.1108/02635570310497675](https://doi.org/10.1108/02635570310497675).
- Hayes, A.F. (2017), *Introduction To Mediation, Moderation, and Conditional Process Analysis: A Regression-Based Approach*, Guilford publications, New York.
- Ho, S.S., Chuah, A.S.F., Kim, N. and Tandoc Jr, E.C. (2022), "Fake news, real risks: how online discussion and sources of fact-check influence public risk perceptions toward nuclear energy", *Risk Analysis*, Vol. 42 No. 11, pp. 2569-2583, doi: [10.1111/risa.13980](https://doi.org/10.1111/risa.13980).
- Howard, M.C. and Henderson, J. (2023), "A review of exploratory factor analysis in tourism and hospitality research: identifying current practices and avenues for improvement", *Journal of Business Research*, Vol. 154, 113328, doi: [10.1016/j.jbusres.2022.113328](https://doi.org/10.1016/j.jbusres.2022.113328).
- Jiang, G., Liu, F., Liu, W., Liu, S., Chen, Y. and Xu, D. (2021), "Effects of information quality on information adoption on social media review platforms: moderating role of perceived risk", *Data Science and Management*, Vol. 1 No. 1, pp. 13-22, doi: [10.1016/j.dsm.2021.02.004](https://doi.org/10.1016/j.dsm.2021.02.004).
- Jin, E. and Atkinson, L. (2021), "The moderating role of emotion: the combinatory effects of positive emotion and news framing techniques on climate change attitudes", *Journalism and Mass Communication Quarterly*, Vol. 98 No. 3, pp. 749-768, doi: [10.1177/1077699020988105](https://doi.org/10.1177/1077699020988105).
- Kasperson, R.E., Renn, O., Slovic, P., Brown, H.S., Emel, J., Goble, R., Kasperson, J.X. and Ratick, S. (1988), "The social amplification of risk: a conceptual framework", *Risk Analysis*, Vol. 8 No. 2, pp. 177-187, doi: [10.1111/j.1539-6924.1988.tb01168.x](https://doi.org/10.1111/j.1539-6924.1988.tb01168.x).
- Kasperson, R.E., Webler, T., Ram, B. and Sutton, J. (2022), "The social amplification of risk framework: new perspectives", *Risk Analysis*, Vol. 42 No. 7, pp. 1367-1380, doi: [10.1111/risa.13926](https://doi.org/10.1111/risa.13926).
- Kim, A. and Dennis, A.R. (2019), "Says who? The effects of presentation format and source rating on fake news in social media", *MIS Quarterly*, Vol. 43 No. 3, pp. 1025-1039, doi: [10.25300/misq/2019/15188](https://doi.org/10.25300/misq/2019/15188).
- Kim, D.J., Ferrin, D.L. and Rao, H.R. (2008), "A trust-based consumer decision-making model in electronic commerce: the role of trust, perceived risk, and their antecedents", *Decision Support Systems*, Vol. 44 No. 2, pp. 544-564, doi: [10.1016/j.dss.2007.07.001](https://doi.org/10.1016/j.dss.2007.07.001).
- Kim, A., Moravec, P.L. and Dennis, A.R. (2019), "Combating fake news on social media with source ratings: the effects of user and expert reputation ratings", *Journal of Management Information Systems*, Vol. 36 No. 3, pp. 931-968, doi: [10.1080/07421222.2019.1628921](https://doi.org/10.1080/07421222.2019.1628921).
- Kim, A., Moravec, P.L. and Dennis, A.R. (2023), "When do details matter? News source evaluation summaries and details against misinformation on social media", *International Journal of Information Management*, Vol. 72, 102666, doi: [10.1016/j.ijinfomgt.2023.102666](https://doi.org/10.1016/j.ijinfomgt.2023.102666).
- Koller, M. (1988), "Risk as a determinant of trust", *Basic and Applied Social Psychology*, Vol. 9 No. 4, pp. 265-276, doi: [10.1207/s15324834baspp0904_2](https://doi.org/10.1207/s15324834baspp0904_2).
- Kukar-Kinney, M. and Walters, R.G. (2003), "Consumer perceptions of refund depth and competitive scope in price-matching guarantees: effects on store patronage", *Journal of Retailing*, Vol. 79 No. 3, pp. 153-160, doi: [10.1016/s0022-4359\(03\)00038-1](https://doi.org/10.1016/s0022-4359(03)00038-1).
- Kumar, A., Shankar, A., Behl, A., Arya, V. and Gupta, N. (2023), "Should I share it? Factors influencing fake news-sharing behaviour: a behavioural reasoning theory perspective", *Technological Forecasting and Social Change*, Vol. 193, 122647, doi: [10.1016/j.techfore.2023.122647](https://doi.org/10.1016/j.techfore.2023.122647).

- Lee, S. and Kim, K. (2023), "Perceived influence of partisan news and online news participation: third-person effect, hostile media phenomenon, and cognitive elaboration", *Communication Research*, Vol. 50 No. 7, pp. 854-878, doi: [10.1177/00936502221127494](https://doi.org/10.1177/00936502221127494).
- Lin, X. and Wang, X. (2020), "Examining gender differences in people's information-sharing decisions on social networking sites", *International Journal of Information Management*, Vol. 50, pp. 45-56, doi: [10.1016/j.ijinfomgt.2019.05.004](https://doi.org/10.1016/j.ijinfomgt.2019.05.004).
- Majerczak, P. and Strzelecki, A. (2022), "Trust, media credibility, social ties, and the intention to share towards information verification in an age of fake news", *Behavioral Sciences*, Vol. 12 No. 51, pp. 1-17, doi: [10.3390/bs12020051](https://doi.org/10.3390/bs12020051).
- Masuda, J.R. and Garvin, T. (2006), "Place, culture, and the social amplification of risk", *Risk Analysis*, Vol. 26 No. 2, pp. 437-454, doi: [10.1111/j.1539-6924.2006.00749.x](https://doi.org/10.1111/j.1539-6924.2006.00749.x).
- Mitchell, V.W. (1999), "Consumer perceived risk: conceptualisations and models", *European Journal of Marketing*, Vol. 33 No 1/2, pp. 163-195, doi: [10.1108/03090569910249229](https://doi.org/10.1108/03090569910249229).
- Narasimhan, R. and Jayaram, J. (1998), "Causal linkages in supply chain management: an exploratory study of North American manufacturing firms", *Decision Sciences*, Vol. 29 No. 3, pp. 579-605, doi: [10.1111/j.1540-5915.1998.tb01355.x](https://doi.org/10.1111/j.1540-5915.1998.tb01355.x).
- Nicolaou, A.I. and McKnight, D.H. (2006), "Perceived information quality in data exchanges: effects on risk, trust, and intention to use", *Information Systems Research*, Vol. 17 No. 4, pp. 332-351, doi: [10.1287/isre.1060.0103](https://doi.org/10.1287/isre.1060.0103).
- Nooteboom, B., Berger, H. and Noorderhaven, N.G. (1997), "Effects of trust and governance on relational risk", *Academy of Management Journal*, Vol. 40 No. 2, pp. 308-338, doi: [10.2307/256885](https://doi.org/10.2307/256885).
- Omar, B., Apuke, O.D. and Nor, Z.M. (2024), "The intrinsic and extrinsic factors predicting fake news sharing among social media users: the moderating role of fake news awareness", *Current Psychology*, Vol. 43 No. 2, pp. 1235-1247, doi: [10.1007/s12144-023-04343-4](https://doi.org/10.1007/s12144-023-04343-4).
- Ortega Egea, J.M. and Román González, M.V. (2011), "Explaining physicians' acceptance of EHCR systems: an extension of TAM with trust and risk factors", *Computers in Human Behavior*, Vol. 27 No. 1, pp. 319-332, doi: [10.1016/j.chb.2010.08.010](https://doi.org/10.1016/j.chb.2010.08.010).
- Pennycook, G. and Rand, D.G. (2019), "Lazy, not biased: susceptibility to partisan fake news is better explained by lack of reasoning than by motivated reasoning", *Cognition*, Vol. 188, pp. 39-50, doi: [10.1016/j.cognition.2018.06.011](https://doi.org/10.1016/j.cognition.2018.06.011).
- Podsakoff, P.M. and Organ, D.W. (1986), "Self-reports in organizational research: problems and prospects", *Journal of Management*, Vol. 12 No. 4, pp. 531-544, doi: [10.1177/014920638601200408](https://doi.org/10.1177/014920638601200408).
- Podsakoff, P.M., MacKenzie, S.B., Lee, J.-Y. and Podsakoff, N.P. (2003), "Common method biases in behavioral research: a critical review of the literature and recommended remedies", *Journal of Applied Psychology*, Vol. 88 No. 5, pp. 879-903, doi: [10.1037/0021-9010.88.5.879](https://doi.org/10.1037/0021-9010.88.5.879).
- Roozenbeek, J. and van der Linden, S. (2019), "The fake news game: actively inoculating against the risk of misinformation", *Journal of Risk Research*, Vol. 22 No. 5, pp. 570-580, doi: [10.1080/13669877.2018.1443491](https://doi.org/10.1080/13669877.2018.1443491).
- Schaewitz, L., Kluck, J.P., Klösters, L. and Krämer, N.C. (2020), "When is disinformation (in)credible? Experimental findings on message characteristics and individual differences", *Mass Communication and Society*, Vol. 23 No. 4, pp. 484-509, doi: [10.1080/15205436.2020.1716983](https://doi.org/10.1080/15205436.2020.1716983).
- Schneider, C.R., Dryhurst, S., Kerr, J., Freeman, A.L.J., Recchia, G., Spiegelhalter, D. and van der Linden, S. (2021), "COVID-19 risk perception: a longitudinal analysis of its predictors and associations with health protective behaviours in the United Kingdom", *Journal of Risk Research*, Vol. 24 No 3-4, pp. 294-313, doi: [10.1080/13669877.2021.1890637](https://doi.org/10.1080/13669877.2021.1890637).
- Seyd, B. and Bu, F. (2022), "Perceived risk crowds out trust? Trust and public compliance with coronavirus restrictions over the course of the pandemic", *European Political Science Review*, Vol. 14 No. 2, pp. 155-170, doi: [10.1017/s1755773922000078](https://doi.org/10.1017/s1755773922000078).

- Shen, G. C.-C., Chiou, J.-S., Hsiao, C.-H., Wang, C.-H. and Li, H.-N. (2016), "Effective marketing communication via social networking site: the moderating role of the social tie", *Journal of Business Research*, Vol. 69 No. 6, pp. 2265-2270, doi: [10.1016/j.jbusres.2015.12.040](https://doi.org/10.1016/j.jbusres.2015.12.040).
- Shin, N. and Cheng, T. (2023), "Gaining user confidence in banking industry's quest for digital transformation: a product-service system management perspective", *Industrial Management and Data Systems*, Vol. 123 No. 8, pp. 2216-2240, doi: [10.1108/imds-06-2022-0358](https://doi.org/10.1108/imds-06-2022-0358).
- Shin, J., Lewis, S.C., Kim, S. and Thorson, K. (2024), "Does high-quality news attract engagement on social media? Mediatization, media logic, and the contrasting values that shape news sharing, liking, and commenting on Facebook", *New Media and Society*, Vol. ahead-of-print No. ahead-of-print, 14614448241228851, doi: [10.1177/14614448241228851](https://doi.org/10.1177/14614448241228851).
- Thompson, N., Wang, X. and Daya, P. (2020), "Determinants of news sharing behavior on social media", *Journal of Computer Information Systems*, Vol. 60 No. 6, pp. 593-601, doi: [10.1080/08874417.2019.1566803](https://doi.org/10.1080/08874417.2019.1566803).
- Torres, R., Gerhart, N. and Negahban, A. (2018), "Epistemology in the era of fake news: an exploration of information verification behaviors among social networking site users", *SIGMIS Database*, Vol. 49 No. 3, pp. 78-97, doi: [10.1145/3242734.3242740](https://doi.org/10.1145/3242734.3242740).
- Wang, S., Pang, M.-S. and Pavlou, P.A. (2022), "Seeing is believing? How including a video in fake news influences users' reporting of fake news to social media platforms", *MIS Quarterly*, Vol. 46 No. 3, pp. 1323-1354, doi: [10.25300/misq/2022/16296](https://doi.org/10.25300/misq/2022/16296).
- Wirz, C.D., Xenos, M.A., Brossard, D., Scheufele, D., Chung, J.H. and Massarani, L. (2018), "Rethinking social amplification of risk: social media and Zika in three languages", *Risk Analysis*, Vol. 38 No. 12, pp. 2599-2624, doi: [10.1111/risa.13228](https://doi.org/10.1111/risa.13228).
- Wu, Y., Ngai, E.W.T., Wu, P. and Wu, C. (2022), "Fake news on the internet: a literature review, synthesis and directions for future research", *Internet Research*, Vol. 32 No. 5, pp. 1662-1699, doi: [10.1108/intr-05-2021-0294](https://doi.org/10.1108/intr-05-2021-0294).
- Yang, Y., Liu, Y., Li, H. and Yu, B. (2015), "Understanding perceived risks in mobile payment acceptance", *Industrial Management and Data Systems*, Vol. 115 No. 2, pp. 253-269, doi: [10.1108/imds-08-2014-0243](https://doi.org/10.1108/imds-08-2014-0243).
- Yi, M.Y., Yoon, J.J., Davis, J.M. and Lee, T. (2013), "Untangling the antecedents of initial trust in Web-based health information: the roles of argument quality, source expertise, and user perceptions of information quality and risk", *Decision Support Systems*, Vol. 55 No. 1, pp. 284-295, doi: [10.1016/j.dss.2013.01.029](https://doi.org/10.1016/j.dss.2013.01.029).
- Zhang, X.A. and Cozma, R. (2022), "Risk sharing on Twitter: social amplification and attenuation of risk in the early stages of the COVID-19 pandemic", *Computers in Human Behavior*, Vol. 126, 106983, doi: [10.1016/j.chb.2021.106983](https://doi.org/10.1016/j.chb.2021.106983).
- Zheng, Y., Zhao, K. and Stylianou, A. (2013), "The impacts of information quality and system quality on users' continuance intention in information-exchange virtual communities: an empirical investigation", *Decision Support Systems*, Vol. 56, pp. 513-524, doi: [10.1016/j.dss.2012.11.008](https://doi.org/10.1016/j.dss.2012.11.008).
- Zhou, T., Li, H. and Liu, Y. (2010), "The effect of flow experience on mobile SNS users' loyalty", *Industrial Management and Data Systems*, Vol. 110 No. 6, pp. 930-946, doi: [10.1108/02635571011055126](https://doi.org/10.1108/02635571011055126).
- Zhou, Z., Li, T., Liu, C., Zhou, Y., Li, P. and Wen, S. (2023), "Why do social media users follow tourism-related posts? Roles of bloggers and posts in trip planning", *Industrial Management and Data Systems*, Vol. 123 No. 12, pp. 3080-3108, doi: [10.1108/imds-11-2022-0692](https://doi.org/10.1108/imds-11-2022-0692).

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